

Miniproject_1

Deadline: Nov. 5th

You are a Bioengineer working on an applied data science project for a Big Pharma company that commercializes solutions for diabetes (diagnostics, drug products,...).

In collaboration with hospitals and the public health system, the company is working on finding ways to predict the likely evolution of the condition with time, in order to anticipate needs and recommend corrective actions.

You have been asked to explore ways to predict the 1-year diabetes progression based on the following clinical parameters, which are regularly monitored for diabetes patients:

- age — age in years
- sex — nominally binary encoded as -1 / +1
- bmi — body mass index
- bp — average blood pressure
- s1 — tc: total serum cholesterol
- s2 — ldl: low-density lipoproteins
- s3 — hdl: high-density lipoproteins
- s4 — tch: total cholesterol / HDL ratio
- s5 — ltg: possibly log of serum triglycerides
- s6 — glu: blood glucose level

Your company has reached an agreement with the scikit-learn developers to include a convenient dataset as part of `sklearn.datasets`.

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Follow the guidelines of Aurélien Géron's HoML book to build a full ML pipeline to predict 1-year diabetes progression from clinical variables.

Replicate the flow from the California housing example:

EDA → stratified train/test split → preprocessing and modeling with `ColumnTransformer`/'Pipeline' → model comparison and cross-validation → hyperparameter search → final test evaluation → error analysis and model export.

Deliverables: A notebook including

- EDA: basic distributions, correlations, and scatterplots of relevant variables.
- Data splitting: train/test with stratification using a key variable (use `bmi` in quantiles, analogous to the income-based stratification in the housing example).
- Pipelines: use `ColumnTransformer`/'Pipeline' to include imputation (for consistency), scaling, and (in one variant) polynomial/interaction features.
- Training & validation: compare several models with `cross\_val\_score` ('LinearRegression', 'Ridge', 'Lasso', 'ElasticNet', 'RandomForestRegressor', 'SVR').
- Hyperparameter tuning: `GridSearchCV` for at least two model families (one linear regularized and one non-linear).
- Final evaluation on the held-out test set with `RMSE`, `MAE`, and `R²`.
- Error analysis and, if applicable, feature importances / coefficients.
- Model export with `joblib`.
- A short write-up (a Markdown cell in the notebook) (5–10 lines) summarizing results, chosen model, and suggested improvements.

Grading:

- Faithful reproduction of the overall flow.
- Clear EDA and justified stratification.
- Sound pipelines and cross-validation.
- Reasonable hyperparameters & metrics.
- Insightful interpretation and communication.